Q2.4 A heat exchanger heats the process fluid from 25°C to 60°C using hot water entering at 80°C and leaving at 65°C. For the maximum flow rate of 6 L/min, calculate the required heat-transfer area using the LMTD method and (U) = 500 W/m²K. Also briefly state one effect of 20% fouling on the required area.

ANSWER Counter-current arrangement is assumed.

$Q = m C_p \Delta T$ (where m = mass flow rate)

$= \rho \times Q \times C_p(\text{water}) \times \Delta T_{fluid}$

$= 1000 \times 6 \times 10^{-3}/60 \times 4180 \times (60-25)$

$= 14.63$ kW

$\Delta T_1 = T(\text{hot, in}) - T(\text{cold, out}) = 80 - 60 = 20°C$

$\Delta T_2 = T(\text{hot, out}) - T(\text{cold, in}) = 65 - 25 = 40°C$

$LMTD = (\Delta T_2 - \Delta T_1) / (\ln \Delta T_2 - \ln \Delta T_1) = 20/\ln 2 = 28.85°C$

$A = Q/(U \times LMTD) = 14630/(500 \times 28.85) = 1.014\ m^2$

Fouling adds a resistance in series with the clean overall coefficient: $1/U_{dirty} = 1/U_{clean} + R_f$. So 20% fouling is equivalent to saying $U_{dirty} = U_{clean} \times 0.8$.

$\Rightarrow A_{dirty} = 14630/(500 \times 28.85 \times 0.8) = 1.268.$

Nearly 25% larger heat-transfer area is needed to deliver the same duty since fouling reduces the overall heat-transfer coefficient. If the physical exchanger area is kept fixed, fouling instead causes the outlet temperature to fall short of the target unless compensated by increasing flow, temperature or cleaning frequency.

TASK 3 SENSOR ENGINEERING AND CALIBRATION

Pressure measurement range: 0–4 bar, with sensor output 4–20 mA; temperature measurement range: 0–100°C; ADC: 12-bit.

Q3.1 Calculate pressure sensor resolution (write formula) and explain whether this is sufficient for ±1% control.

Resolution = Full-Scale Span / 2^n

$= 4 \text{ bar} / 2^{12}$

$= 0.977 \text{ mbar}$ per bit

Check: ±1% of 4 bar span = ±40 mbar of allowed measurement error. The ADC's quantisation step is ≈40.96× finer than the tolerance band, so the existing setup is good enough for ±1% control.

Q3.2 A temperature sensor has a range of 0–100°C and output 4–20 mA. Plot the calibration graph (temperature vs. current) and determine the temperature at 12 mA.

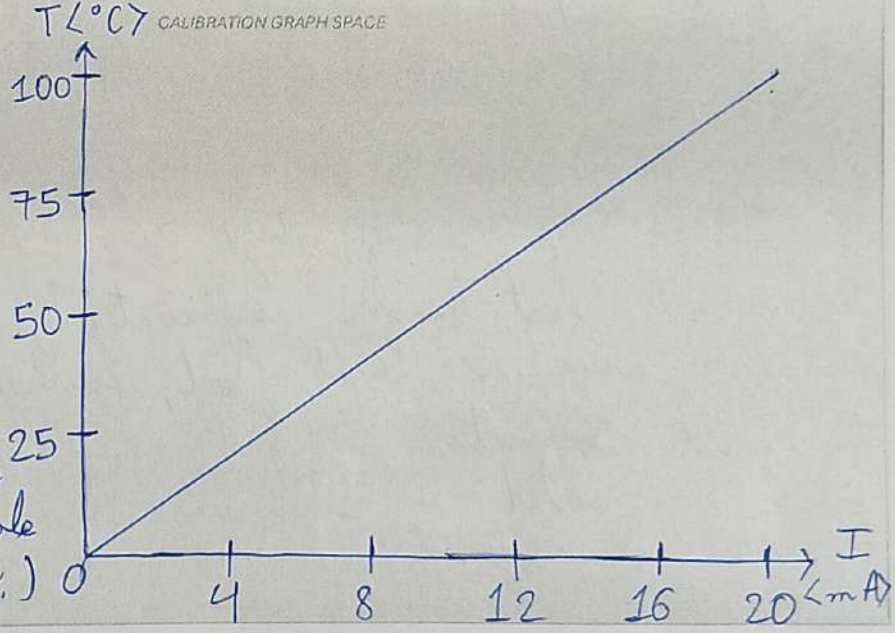
$T = \frac{I-4}{20-4} \times 100$

$= 6.25(I-4)$

At $I = 12\text{mA}$:

$T = 50°C$

(electrical mid-scale and temp^{r} mid-scale coincide – linearity.)



Q3.3 Design a suitable 4–20 mA to 0–3.3 V signal-conditioning circuit. Calculate the required shunt resistor. Also explain why signal conditioning is required between the temperature transmitter and controller.

ANSWER $R_{SH} = V_{FS} / I_{FS} = 3.3 / 0.02 = 165\ \Omega$

The 4-20mA current loop is chosen for transmission because current gives strong noise immunity over long cable runs due to no attenuation with cable length, but MCUs only accept voltage. Signal conditioning performs current-to-voltage conversion, level shifting/scaling into the ADC's input window, filtering of induced EMI/noise, and protection of the sensitive digital input from field-side transients, fault currents or miswiring.

Q3.4 If the sensor drifts after 120 days to +0.08 bar, develop: calibration SOP, zero correction method, field validation procedure.

ANSWER Drift rate = 0.08 bar / 120 d = 0.67 mbar/day

a) SOP: Isolate the transmitter and vent it to atmosphere; record the as-found zero (+0.08 bar). Check span at 0, 25, 50, 75, 100% of the 0–4 bar range against a traceable reference (dead-weight tester). Adjust zero/span, record as-left values, issue a calibration certificate.

b) Zero correction: Apply a −0.08 bar trim offset (digital trim or physical screw); corrected reading = raw reading − 0.08 bar

c) Field validation: Re-check the corrected transmitter at 0, 50, 100% of span against a portable reference, confirm the DCS/PLC shows the corrected value end-to-end, and set the next due date from the drift rate.

Q3.5 You have a ½-inch pipe with air pressure of 15 bar in it and an Aira Euro SOV connected to the pipe, running on 24 VDC with 8 W. The outlet pressure of this SOV is unstable and the coil gets heated up (SOV maximum rating is 20 bar, max temp 150°C). What should be done? (Answers can be structured as case-1, case-2, etc.)

ANSWER $I = P/V_{DC} = 8/24 = 0.333\,A$; $R = V/I = 24/0.333 = 72\,\Omega$

Since 15 bar < 20 bar rating, the SOV is not over-pressured.

Case I: Coil electrical fault/duty-cycle overload — Measure coil current/resistance against the ratings. A shorted turn lowers resistance and raises current and heat, and the same goes for sustained over-voltage. ~~or~~ Remedy – Verify supply voltage, confirm duty-cycle rating matches usage, replace coil if resistance is off-spec.

Case II: Mechanical/flow-path fault causing chatter — Debris in the orifice or a damaged plunger can prevent full seating, causing rapid open/close chatter – destabilising outlet pressure and raising avg. coil current via repeated inrush. Remedy – Inspect valve internals, clean/replace the plunger, fit an upstream strainer in case of debris.

Case III: Insufficient min. operating pressure differential — Below the manufacturer's minimum, pilot-operated valves need a min. inlet/outlet ΔP to fully open else the valve may hunt between partially open/closed states, giving unstable outlet pressure. Remedy – Check op. ΔP against datasheet's min. and adjust upstream setup if too low.

Case IV: Poor heat dissipation/ambient conditions — A poorly ventilated coil can run close to its 150°C limit even at rated ~~power~~ power. Remedy – Improve spacing, add a heat sink, monitor coil and de-rate/replace if needed.